CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 19: Constructions (Circles)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write the steps to construct a tangent to a circle at a point P on it.

2. Write the steps to construct the two tangents to a circle from a point P outside it.

3. A point is 13 cm from the centre of a circle of radius 5 cm. Calculate the length of each tangent from it.

4. Which lines are drawn to find the circumcentre of a triangle? Which are drawn to find the incentre?

5. A triangle has sides 5 cm, 12 cm and 13 cm. Find its circumradius.

6. For the triangle in the previous question, find the inradius.

7. An equilateral triangle has side 9 cm. Find its circumradius and its inradius.

8. Where does the circumcentre of an obtuse-angled triangle lie?

9. A regular hexagon has side 5 cm. Find the radius of its circumscribed circle and of its inscribed circle.

10. Write the steps to construct the incircle of a triangle ABC.

MathSetu

Answer Key

1. Join OP; construct the perpendicular to OP at P.
2. Bisect OP at M; draw the circle with centre M and radius MO to cut the given circle at A, B; join PA, PB.
3. 12 cm
4. Perpendicular bisectors of the sides; bisectors of the angles
5. It is right-angled; circumradius $= 6.5$ cm
6. Area 30 cm², $s = 15$ cm; $r = 2$ cm
7. $3\sqrt{3} \approx 5.2$ cm and $\frac{3\sqrt{3}}{2} \approx 2.6$ cm
8. Outside the triangle
9. 5 cm and $\frac{5\sqrt{3}}{2} \approx 4.33$ cm
10. Bisect two angles to meet at I; draw $ID \perp BC$; draw the circle with centre I and radius ID.

MathSetu